Pacgate AI - Two-AIPC Deployment Handbook

> Clone the repo on each machine, run the same install steps, and both machines become fully operational with deer-flow research and qm collaboration. Targets release 0.1.28 - handbook updated 2026-10-13 Chinese version: AIPC-DEPLOYMENT-HANDBOOK-ZH.md Prerequisites: Docker Desktop, Ollama, Node.js 24+. `install.ps1` pulls the models listed in `ollama-models.txt`. PowerShell 7 (`pwsh`) is optional - the installer runs on the built-in Windows PowerShell 5.1 as well. (Before 2026-10-11 that claim was false, in two ways: PS 5.1 turns any native command that writes to stderr into a terminating error when a redirection is attached, and `docker compose up -d` writes progress to stderr on every run, so the installer aborted before bootstrapping the tenant and admin; and the admin-registration call used the PowerShell 7-only `-SkipHttpErrorCheck` parameter, which 5.1 rejects before sending the request. `install.ps1`, `setup-qm.ps1` and `smoke-deer-flow-outputs.ps1` now route every stderr-writing call through `Invoke-WithNativeStderr` and make API calls through `Invoke-ApiRequest`. See `deploy/DEFECT-ps51-native-stderr.md`.)

⚠ Significant findings (2026-09-02) — read before deploying AIPC #2

These were discovered during the AIPC #1 pilot and are already fixed in this repo. AIPC #2 must pull the updated code (see Stage 1) so it gets these fixes.

> Update 2026-09-23 (supersedes the 2026-09-15 "clone either" note). The two repos are NO LONGER identical — `pacgate-ai/pacgate-ai-pr` is 26 commits behind and missing the workflow-wiring fixes (`b7fc540`, `039afdc`), so a fork clone serves 10 built-in workflows instead of the firm's 222, with no error shown anywhere. Clone `JZKK720/pacgate-ai-pr` (see Stage 1). Which repo you push a release tag to still decides the GHCR namespace — see `plans/012-master-release-namespace.md`.

1. deer-flow agent could not query pacgate's legal databases. Root cause: no tool was wired to pacgate-api's `/api/kb/search` (RAG) or `/api/search` (legal connectors), and the memory adapter silently fell back to local files (`PacgateMemoryStorage requires PACGATE_MATTER_ID`). Fix: a new `pacgate-mcp` service (FastMCP) exposes `pacgate_kb_search`, `pacgate_connector_search`, `pacgate_list_connectors` to deer-flow. Registered in `deer-flow-extensions-config.json` beside openviking.
2. openviking MCP had the wrong API key baked in. `deer-flow-extensions-config.json` used the app key (`OPENVIKING_API_KEY`) but openviking's `root_api_key` is `OPENVIKING_ROOT_API_KEY`. A wrong key made openviking return 401, which rolled back the entire MCP tool load (deer-flow uses `asyncio.gather`), so no MCP tools appeared. Fix: use `OPENVIKING_ROOT_API_KEY` (template now uses `${OPENVIKING_ROOT_API_KEY}`).
3. Prefer `docker compose restart deer-flow` over `--force-recreate` for config changes. `.deer-flow` (SQLite DB, admin user, threads, `.jwt_secret`) is now bind-mounted at `./data/deer-flow:/app/backend/.deer-flow` in BOTH `compose.prod.yaml` and `compose.bundle.yaml`, so a recreate no longer loses it. Measured 2026-10-28: a `--force-recreate deer-flow` on the dev box left the DB, the `10/21 .jwt_secret` and the user threads intact. This was previously documented as destructive and that reason is now stale. The guidance still stands for a different reason: a recreate also re-runs `/setup`-adjacent startup work and interrupts in-flight runs, and `patches/*.py` plus `deer-flow-config.yaml` are read at process start, so `restart` is enough to apply them. Recreate only when the compose file itself changed (new mounts, env, or image).
4. QM sign-in needs an email transport, not Outlook SMTP. The old SMTP path (`smtp.office365.com` + app password) is broken — Microsoft retired Basic Auth / app passwords for Exchange Online (Sep 2025). `qm check` failed with `535 5.7.139 Authentication unsuccessful`. Fix: qm's auth broker now supports two transports — Mailpit SMTP catcher (pilot; links land at `http://localhost:8025`, no key needed) and Resend

(production; supply `RESEND_API_KEY`). See Stage 4. (Originally this item said Resend was the only option; Stage 4's two-transport setup supersedes it.)

5. The qm web-ui cannot self-authenticate. Its server (`/app/server/index.ts`) sets `AUTH_MODE = COOKIE_AUTH ? "dev" : "portal"`. Because `CORE_SIGNING_SECRET` is set, it's in portal mode and requires a portal-issued identity token. There is no no-secret way to reach the web-ui directly — you must run `portal+auth` (Resend) or an external OIDC provider. `ADMIN_GRANTS` is an authorization seed, not a sign-in.
6. Git push to `JZKK720/pacgate-ai-pr` was blocked for the `pacgate-ai` account (403, needs 2FA grant). Workaround: the `pacgate-ai` account could fork and push there. All fixes are now merged into both `main` branches, so this is historical — but note the two remotes are separate publishing targets for GHCR (see `plans/012-master-release-namespace.md`).

Architecture: two identical machines

Both AIPCs run the complete stack:

```
Each AIPC machine:
  nginx :8089  -> pacgate-api :8080 (Rust metadata API)
               -> deer-flow  :8001 (research workspace)
  Postgres :5432 (local metadata DB)
  OpenViking :1933 (long-term memory lane, MCP)
  qm :8182 (co-working workspace, runs via `qm up`)
  Ollama :11434 (native, GPU/NPU)
```

Each machine is self-contained and independently operational. Lawyers on either machine can use both research mode (deer-flow at `http://localhost:8089/research/`) and collaboration mode (qm at `http://localhost:8182`) without depending on the other machine.

If you later want shared matter data across both machines, connect them with a private mesh (Tailscale or WireGuard) and decide on a sync or single-authority model. That is a post-pilot decision, not a deployment prerequisite.

What you need before starting

- GitHub access to the source repo — `JZKK720/pacgate-ai-pr` (public; a plain clone needs no auth). Do not clone the `pacgate-ai` fork — it is 26 commits behind and serves 10 workflows instead of 222 (see Stage 1). A PAT or `gh auth login` is only required if you intend to push.
- Docker Desktop running on both AIPCs
- Ollama running on both AIPCs (`install.ps1` pulls the models it needs)
- `ollama signin` completed on each AIPC if the cloud-tagged deepseek models are in use
- Node.js 24+ installed on both AIPCs (for qm)
- No `docker login ghcr.io` needed — the Pacgate runtime images are published as public GHCR packages (see Stage 0).

Stage 0: Runtime images (dev machine, already done)

The runtime is published on GHCR and needs no rebuild on the AIPC.

Current versions (the ones to pull):

Image	Status
ghcr.io/jzkk720/pacgate-api:0.2.33	The sanitizer release. Fixes the verifier that let a document pass while still holding 身份证号, 统一社会信用代码, 律师执业证号, 银行账号, 港澳通行证, 车牌, 微信号 and 纳税人识别号 (see below). Also adds the matter-workspace rollup and two remediation endpoints: POST /api/documents/:id/sanitize-rescan and PATCH /api/documents/:id/sanitize-reset .
ghcr.io/jzkk720/pacgate-mcp:0.2.33	Exposes 24 MCP tools to deer-flow; adds pacgate_rescan_residue and pacgate_reset_sanitization for the remediation path. Note server.py is bind-mounted into the container, so git pull delivers these without a new image.
ghcr.io/jzkk720/deer-flow-pacgate:0.2.33	Unchanged this release.
ghcr.io/jzkk720/deer-flow-frontend-pacgate:0.2.33	Unchanged this release.
ghcr.io/jzkk720/ocr-service:0.2.33	PaddleOCR extraction; first-class since 0.1.16. Its PP-OCRv4 weights are baked into the image from 0.1.28, so extraction needs no external model host and works offline.
ghcr.io/volcengine/openviking@sha256:46f9e34c…	Pinned by digest in compose.prod.yaml . Upstream public image.

0.2.14 is a correctness fix, not a feature release. A client scan of a real scanned contract produced `verdict: pass` with 13 replacements while eight identifier classes stayed verbatim in text that was then indexed as searchable - and the document carried the `sanitized` label, so nothing downstream looked at it again. The recorded reason was `"noise replay: 1 of 1 surviving matches dropped by the same suppressor the redaction used"`: the verifier replayed the redactor's own suppressor, so any survivor it missed on the way in was missed again on the way out.

The verifier now never lets that suppressor touch a rule match. Two capabilities exist because the fix is forward-only - text already mislabelled needs a way back: `pacgate_rescan_residue` re-reads the STORED chunks (rules only, no model) and names what survived, and `pacgate_reset_sanitization` returns a `sanitized` document to `pending` so a re-run is a real rewrite rather than a no-op. Until a machine has been rescanned, do not treat its RAG results as clean.

Namespace and version corrected 2026-09-22. This table previously listed `ghcr.io/pacgate-ai/*` at 0.1.0/0.1.3. `pacgate-ai` is the legacy mirror; the live namespace is `jzkk720` and publishing moved there in plan 016. The old `pacgate-ai/...-frontend-pacgate:0.1.0` row was also labelled "Published" while returning 404 - it never resolved. All five `jzkk720/*` images at the current pin (0.1.28, table above) return HTTP 200 anonymously.

Historical release table (retained for provenance, superseded):

Image	Status
ghcr.io/pacgate-ai/pacgate-api:0.1.3	Published at the time. Fixed the 0.1.1 container-networking bug.
ghcr.io/pacgate-ai/pacgate-mcp:0.1.3	Published at the time.
ghcr.io/pacgate-ai/deer-flow-pacgate:0.1.3	Published at the time.

Image	Status
ghcr.io/pacgate-ai/deer-flow-frontend-pacgate:0.1.0	Never published - 404. Do not use.

All Pacgate packages must be set to public visibility on GHCR so an AIPC can pull without registry credentials. Verify before rollout:

```
# Expect HTTP 200 with no docker login. 401/403 means the package is still private.
# 404 means the tag does not exist in that namespace — usually a drift between the
# pins in compose.prod.yaml and where the images were actually published.
#
# (The Accept header is REQUIRED — omit it and a public manifest returns 404, not 200.)
#
# This reads the four pins straight out of compose.prod.yaml, so it can never go
# stale: previously this snippet hard-coded an old tag that had been removed, and
# reported a healthy system as broken.
$acc = "application/vnd.oci.image.index.v1+json,application/vnd.docker.distribution.manifest.list.v2+json,ap
$compose = "deploy/client-bundle/compose.prod.yaml"
$pins = Select-String -Path $compose -Pattern "image:\s*(ghcr\.io/[^\s]+)" -AllMatches |
        ForEach-Object { $_.Matches } | ForEach-Object { $_.Groups[1].Value } |
        Where-Object { $_ -notmatch "openviking" } | Sort-Object -Unique

foreach ($pin in $pins) {
  $repo = $pin -replace "^ghcr\.io/", ""           # owner/name:tag
  $name = ($repo -split ":")[0]                      # owner/name
  $tag  = ($repo -split ":")[1]
  $t = (Invoke-RestMethod "https://ghcr.io/token?scope=repository:$name`:pull").token
  $code = (Invoke-WebRequest -Uri "https://ghcr.io/v2/$name/manifests/$tag" `
            -Headers @{Authorization="Bearer $t"; Accept=$acc} `
            -Method Head -UseBasicParsing).StatusCode
  "{0,-58} => HTTP {1}" -f $pin, $code
}
```

To flip it (GitHub web UI — the API route 404s for personal accounts): GitHub → your profile → Packages → `pacgate-api` → Package settings → Visibility → Public → Save. Repeat for `deer-flow-pacgate`. This is safe: the images contain only the compiled binary and SQL migrations, every secret is injected at runtime via `.env`, and the installer already has full source access to the same code.

Pruning old image versions

Six packages have accumulated roughly 210 tags, most of them long superseded. Only the newest three releases need to exist for a client to install or roll back.

```
# Show the plan first. This is the default and deletes nothing.
.\scripts\prune-ghcr-versions.ps1

# Then, with a token that has read:packages + delete:packages:
$env:GHCR_PRUNE_TOKEN = '<token>'
.\scripts\prune-ghcr-versions.ps1 -Execute
```

Three things make this harder than it looks, and the script exists because of them:

- The registry API cannot delete GHCR manifests. `DELETE /v2/<repo>/manifests/<digest>` answers 405 for every credential. Deletion goes through the GitHub Packages REST API and needs a token with `read:packages` and `delete:packages`. An ordinary `repo` token gets `403`. A `200` from `ghcr.io/token?...:delete` is not evidence of permission — it is returned regardless of what the token may do.
- A GitHub Actions workflow cannot do this either, and that is structural. `jzkk720` is a user account, not an organisation, so the packages it owns live under `/user/packages/...`. The automatic `GITHUB_TOKEN` is an installation token: it can never act as the user, and its `/repos/{owner}/{repo}/packages` path only sees packages linked to the repo. Measured 2026-11-03 from inside Actions with `permissions: packages: write` — the repo list, the single package, its versions, and the `/orgs/jzkk720/...` variant all returned 404.

An earlier version of this handbook suggested a workflow; it was removed because it could never succeed. A user-scoped token with `delete:packages` is the only route.

- `latest` and a release tag can share one digest. Deleting the version behind an old tag can take `latest` with it. The plan protects every digest that a kept tag or `latest` points at, and says so instead of deleting.
- One digest can carry several tags. `0.2.13` and the leftover `tmp-del` tag point at the same manifest, so the plan collapses them into one delete. Listing them separately would make the second DELETE answer 404 and report a successful run as a failure.

Deletion is irreversible, so the script is dry-run by default, needs `-Execute` plus a typed confirmation, and is covered by `scripts/test-prune-ghcr-versions.ps1` (offline fixtures, no credential) which asserts it cannot drop a kept release or misreport itself.

Keep at least the version you are rolling back to. A client on `0.2.16` that needs to revert should find `0.2.15` still pullable.

Only rebuild and push if the Rust source changes, from the dev machine. Take the tag from the compose pin rather than typing a version — a hard-coded tag here has gone stale twice, and one stale copy already reported a healthy system as broken:

```
cd c:\Users\cubecloud-io\github-pr\pacgate-ai-pr
# Pattern reads the CURRENT namespace (jzkk720). The old form matched
# 'pacgate-ai/pacgate-api' and no longer matches any line in compose.prod.yaml,
# so $tag came back EMPTY and the build/push below silently used a blank tag.
$tag = (Select-String -Path deploy/client-bundle/compose.prod.yaml `
        -Pattern 'jzkk720/pacgate-api:(\S+)').Matches.Groups[1].Value
if (-not $tag) { throw 'could not read the image tag from compose.prod.yaml' }
docker build -t ghcr.io/jzkk720/pacgate-api:$tag -f pacgate-ai/Dockerfile ./pacgate-ai
docker push  ghcr.io/jzkk720/pacgate-api:$tag
```

In practice prefer the `build-ghcr.yml` workflow so all four images stay in step — see `deploy/README-BUILD.md` and `plans/012-master-release-namespace.md`.

Do not rebuild on the AIPC — the pilot runs the published digests.

> Port note: the stack binds nginx to host port `8089` (the value committed in `deploy/client-bundle/compose.prod.yaml`). If that port is already in use on the machine, edit the `ports:` entry for `nginx` and use the new port in all verification URLs below.

Stage 1: Clone the repo on each AIPC

On both machines:

```
cd C:\
git clone https://github.com/JZKK720/pacgate-ai-pr.git
cd pacgate-ai-pr
git remote -v   # origin MUST be JZKK720/pacgate-ai-pr
```

> AIPC #2 note (CORRECTED 2026-09-23): the two repos are NO LONGER IDENTICAL, so "either clone works" is no longer true. Clone `JZKK720`.
>
> `pacgate-ai/pacgate-ai-pr` is 26 commits behind (last checked 2026-09-23) and is missing `b7fc540` and `039afdc`, so it still carries the original workflow-wiring defect. The 15 workflow YAMLs are present in the fork but are not wired into `pacgate-api`, so the API serves 10 built-in workflows instead of the firm's 222 — with no error shown anywhere. The identity claim above was accurate when written; the divergence came afterwards, which is exactly why a doc must not assert two things are in sync without a check that can fail.
>
> The namespace difference is unchanged — which repo you push a release tag to decides which GHCR namespace the images publish into. See `plans/012-master-release-namespace.md`.

> Cloning needs no credentials now that the repos are public; a PAT or `gh auth login` is only required to push.

If the repo is private and GitHub prompts for credentials, use a personal access token or the GitHub CLI (`gh auth login`).

Stage 2: Deploy the core stack (both machines, identical steps)

Run these steps on each AIPC. The Docker Compose stack starts pacgate-api, Postgres, nginx, and deer-flow.

```
cd C:\pacgate-ai-pr\deploy\client-bundle
copy .env.example .env
notepad .env
```

Fill in these values:

```
PACGATE_DB_PASSWORD=<generate a strong password>
PACGATE_JWT_SECRET=<generate a random hex string>
PACGATE_TENANT_ID=pacgate-law
OPENVIKING_ROOT_API_KEY=<generate a 32-char hex string>
OPENVIKING_API_KEY=<generate a 32-char hex string>
```

`OPENVIKING_API_KEY` is required — the installer renders `deer-flow-extensions-config.json` from it and stops with an error if it is missing or left as `change-me`.

Generate secrets if you need them:

```
# DB password (16 hex)
-join ((1..16) | ForEach-Object { '{0:x}' -f (Get-Random -Maximum 16) })

# JWT secret (32 hex)
-join ((1..32) | ForEach-Object { '{0:x}' -f (Get-Random -Maximum 16) })

# OpenViking keys (32 hex each) — generate a fresh value per line
-join ((1..32) | ForEach-Object { '{0:x}' -f (Get-Random -Maximum 16) })
```

Run the installer:

```
.\install.ps1
```

The installer pulls the (public, no-login) GHCR images, renders `OPENVIKING_CONF_CONTENT` and `deer-flow-extensions-config.json` from the `.env` secrets, starts the Docker Compose stack, and pulls the Ollama models listed in `ollama-models.txt`. If models are already pulled, this step is fast.

Verify the core stack:

```
docker compose -f compose.prod.yaml ps
curl http://localhost:8089/version
curl http://localhost:8089/pacgate/health
```

Expected: all eight containers running (pacgate-db, pacgate-api, deer-flow, deer-flow-frontend, pacgate-mcp, ocr-service, openviking, nginx); `/version` returns `{"version":"0.1.28","revision":"<git sha>"}` and `/pacgate/health` returns `ok`.

> Do not probe `/health` at the nginx root. nginx routes `/` to the deer-flow frontend by design, so `curl http://localhost:8089/health` returns the frontend's 404 page - that reads as a failure but is correct routing. `/version` is also at the root (mapped to the API's `/build-info`); only `/pacgate/*` paths reach the API.

Stage 3: Seed the tenant and provision accounts (both machines)

> `install.ps1` now does this automatically (step 6a). On a current install you do not need to run anything on this page - it is kept for recovery, and the commands below are what the installer runs. Check the install output for

`[OK] tenant 'default-firm' present` and `[OK] admin '...' registered`.

How accounts work since 0.1.22 (read before provisioning people)

- `POST /api/auth/register` is first-user-only: it creates exactly one account on a fresh deployment (the bootstrap admin) and refuses afterwards with 403.
- Every account after the first is created by the admin through `POST /api/auth/users` (Bearer = the admin's token).
- Research workspace sign-in = email + password. Collaboration (qm) sign-in = one-time emailed links allowlisted via `AUTH_ALLOWED_EMAILS`.

On each machine, seed the default tenant, then let the installer bootstrap the admin:

```
# Seed the tenant (idempotent).
#
# THE SLUG MUST MATCH PACGATE_TENANT_ID. It defaults to "default-firm", and the
# registration below looks the tenant up by that slug. An earlier version of this
# page used 'pacgate-law', which does not match, so registration failed with:
#
#   {"error":{"code":"internal_error",
#     "message":"default tenant not found: matter not found: row not found"}}
#
# That reads like a database fault and is really a naming mismatch. If you set
# PACGATE_TENANT_ID to something else in .env, use that value here instead.
docker exec pacgate-db psql -U pacgate -d pacgate -c "INSERT INTO tenants (name, slug) SELECT 'Default Firm'
```

Provision an attorney user (run by the admin once the admin account exists):

```
$login = Invoke-RestMethod -Uri "http://localhost:8089/pacgate/api/auth/login" -Method Post `
  -Body '{"email":"admin@pacgate-law.com","password":"<admin-password>"}' `
  -ContentType "application/json"
$body = @{email="<attorney-email>"; password="<attorney-password>"; role="attorney"} | ConvertTo-Json
Invoke-RestMethod -Uri "http://localhost:8089/pacgate/api/auth/users" -Method Post `
  -Headers @{Authorization="Bearer $($login.token)"} -Body $body -ContentType "application/json"
```

Register the qm bridge service account (the installer does this; manual for recovery). Same admin-provisioned route, same shape as above, with `email="qm-bridge@pacgate-law.com"`.

For the qm collaboration surface: set `AUTH_ALLOWED_EMAILS` (comma-separated) and `ADMIN_GRANTS=<email>:org_admin` in `deploy/client-bundle/qm-pacgate/.env`, then re-run `setup-qm.ps1`.

Stage 3.5: Verify OpenViking memory service (both machines)

OpenViking is the long-term memory lane: deer-flow and qm store conversational context there and recall it in later sessions. It starts as part of the compose stack.

```
curl http://localhost:1933/health
```

Expected: `{"status":"ok","healthy":true,...}`. The installer renders the OpenViking config (Ollama embedding + VLM) into `.env` as `OPENVIKING_CONF_CONTENT` and seeds the server's `ov.conf` on first boot.

Functional check (optional, uses the root key from `.env`):

```
$key = (Get-Content .env | Select-String '^OPENVIKING_ROOT_API_KEY=').Line.Split('=')[1]
$body = '{"jsonrpc":"2.0","id":1,"method":"tools/list"}'
curl.exe -s -X POST http://localhost:1933/mcp -H "X-API-Key: $key" -H "Content-Type: application/json" -H "A
```

Expected: a tool list including `find`, `search`, `read`, `remember`.

Boundary rule: OpenViking stores conversational context only (decisions, preferences, working knowledge). Matter documents and T1-T4-controlled content stay in pacgate-api/pacgate-rag.

Stage 4: Bootstrap qm (both machines, identical steps)

qm runs separately from the Docker Compose stack. Bootstrap it on each machine after the core stack is healthy.

```
cd C:\pacgate-ai-pr\deploy\client-bundle
.\setup-qm.ps1
```

The script prompts for: - Administrator work email (lowercased) - Pacgate bridge email: `qm-bridge@pacgate.local` - Pacgate bridge password: the one you registered in Stage 3

The script generates signing secrets, creates `.env` in the qm-pacgate directory, validates the config with `qm check`, builds the sandbox image with `qm sandbox build`, fetches the static docker CLI the core shells (SHA-256 verified), and builds the `qm-pacgate-sandbox-local` exec-daemon wrapper image.

QM sign-in needs an email transport. The qm auth broker delivers sign-in one-time links. Two supported transports:

- Local/pilot topology (Mailpit SMTP catcher): `SMTP_HOST=mailpit SMTP_PORT=1025 AUTH_EMAIL_TRANSPORT=smtp SMTP_TLS=none`. Links LAND IN THE MAILPIT INBOX — open `http://localhost:8025` to pick up a sign-in link in a pilot. No real email is sent.
- Production topology (Resend): `AUTH_EMAIL_TRANSPORT=resend` with `RESEND_API_KEY` in `deploy/qm-pacgate/.env`, and `AUTH_EMAIL_FROM` set to a Resend-verified sender (an Outlook address is not verified; Microsoft retired Basic Auth/app passwords for Exchange Online, so raw Outlook SMTP is NOT a supported transport):

```
RESEND_API_KEY=re_xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
AUTH_EMAIL_FROM="PacGate <onboarding@resend.dev>"
```

Start qm:

```
cd C:\pacgate-ai-pr\deploy\client-bundle\qm-pacgate
docker compose -f compose.qm.yaml up -d
```

> Use compose, NOT `qm up`. The `@yc-software/qm` CLI's docker lifecycle is POSIX-only - its `which()` shells `/bin/sh`, which does not exist on Windows (measured 2026-10-06: `execFileSync("/bin/sh", ...)` → `ENOENT`), so `qm up`, `qm down`, and `qm status` die with "docker not found on PATH" on every AIPC. `compose.qm.yaml` describes the same topology (same names, volumes, network) and additionally carries the Pacgate patches: `patch/pi-models.ts` and `patch/local-sandbox.ts` are bind-mounted over the core's source, the static docker CLI and the host docker socket are mounted, and the sandbox wrapper image is wired. Those mounts survive every `down`/`up -d` cycle - the model routing and the sandbox lane are durable through compose. Do NOT run `docker compose -f compose.qm.yaml up -d` and a `qm up` against the same directory (they would fight over the same named volumes and network).

Verify qm:

```
# Open http://localhost:8182  (web-ui) — requires a portal identity token
# Open http://localhost:8181  (portal) — the sign-in front door
# Sign in with the admin email (magic link via the configured transport; Mailpit in pilot)
# Send a test message
# Ask: "List available pacgate workflows"
```

> Web-ui auth reality: the qm web-ui (`:8182`) cannot authenticate anyone on its own — it must be reached through the portal (`:8181`), which issues the identity token. If you open `:8182` directly you'll see "reached through the portal." Always go through `:8181`.

> Dev-mode alternative (pilot / single-user): for a local pilot you can run qm in dev/cookie mode instead of the portal. Recreate `qm-pacgate-core` with `NODE_ENV=development` + `ALLOW_UNAUTHENTICATED_CORE=1` and no `CORE_SIGNING_SECRET`, and `qm-pacgate-web-ui` with no `CORE_SIGNING_SECRET`. Then `POST /signin` works directly at `:8182` with `{"user":"<principal>"}` and no Resend key is needed. This is not production-correct (no auth) — use it

> only for a single-user pilot.

Stage 5: Verify deer-flow (both machines)

deer-flow has its OWN login, separate from `pacgate-api`'s. This is the single most common source of "the password is wrong" reports, so read this before trying to sign in.

There are two identity stores and they do not share passwords:

surface	URL	identity store	password comes from
pacgate-api (the API, install.ps1 provisions it)	/pacgate/api/auth/login	Postgres users	.env → PACGATE_API_PASSWORD
deer-flow (the research UI)	/api/v1/auth/*	SQLite data/deer-flow/data/deerflow.db	set at deer-flow's own first-run setup

`PACGATE_API_EMAIL` / `PACGATE_API_PASSWORD` are the pacgate-api principal. They are not deer-flow credentials, and using them against the deer-flow login returns `401 invalid_credentials` - which reads exactly like a wrong password but is a wrong principal. `install.ps1` does not create a deer-flow admin; it cannot, because deer-flow's admin is established by its own `/initialize`.

To sign in to deer-flow:

```
# Open http://localhost:8089/research/
# If it asks you to set up an admin, do that first - that account is deer-flow's own.
# Otherwise sign in with the deer-flow admin you set up previously.
```

If the deer-flow admin password is genuinely lost, it can be re-established without touching anything else. The password format is bcrypt of base64(sha256(password)), written by `app/gateway/auth/password.py`:

```
# 1. Does deer-flow think it is already set up?  (needs_setup=false means an admin exists)
curl.exe -s http://localhost:8089/api/v1/auth/setup-status

# 2. Reset the admin password by writing a fresh hash into deer-flow's OWN store.
#    Run INSIDE the container, against /data/deer-flow/data/deerflow.db.
#    A worked example is in deploy/BENCHMARK-REPORT-2026-10-08.md - it was done there
#    on 2026-10-08 after the first-run passwords were found to be unrecoverable.
#    Change the password after logging in: Settings -> Account.
```

Then verify the research workspace works:

```
# Open http://localhost:8089/research/
# Select or create a matter
# Ask: "Summarize recent force majeure case law in China"
# Verify: response includes citations
# Verify: response is saved to matter memory
```

A useful mechanical check that needs no credentials: a login with a deliberately wrong password must return 401 `invalid_credentials` - not 403 (which means the Origin gate is misconfigured and a browser would be locked out) and not a 5xx (auth service broken).

Stage 5.5: Full-loop operations — deer-flow ↔ QM ↔ OpenViking

Both workspaces share one OpenViking long-term memory lane and one pacgate-api metadata store. This section documents how the two systems talk to each other and how to verify the loop end-to-end.

Topology (verified 2026-09-04)

```
pacgate-ai-bundle_default  (Docker Compose network)
■■■ openviking :1933   ← long-term memory (MCP: find/search/read/remember)
■■■ pacgate-api :8080  ← metadata API (matters/workflows/connectors)
■■■ pacgate-mcp :8000  ← FastMCP bridge exposing pacgate KB/connector search
```

```
■■■ laya :8000         ← optional local decision engine (opt-in profile)
■■■ deer-flow :8001    ← research workspace (consumes openviking + pacgate-mcp)
■■■ nginx :8089        ← ingress (deer-flow frontend + /pacgate/ API)

qm-pacgate  (qm up network)
■■■ qm-pacgate-core    ← co-working agent runtime
■■■ qm-pacgate-web-ui  ← browser chat UI (:8182)
■■■ qm-pacgate-pg      ← qm Postgres

Bridge: qm-pacgate-core is ALSO joined to pacgate-ai-bundle_default, so it can
reach openviking:1933, pacgate-api:8080, and host.docker.internal:11434 (Ollama).
```

How the loop works

1. deer-flow → OpenViking: `deer-flow-extensions-config.json` registers `openviking` as an HTTP MCP server at `http://openviking:1933/mcp` with the root API key (`OPENVIKING_ROOT_API_KEY`). Research runs store and recall conversational context there.
2. deer-flow → pacgate: `pacgate-mcp` (FastMCP) exposes `pacgate_kb_search` / `pacgate_connector_search` / `pacgate_list_connectors` to deer-flow, backed by pacgate-api's RAG + legal connectors.

From 0.2.10 the same server also exposes three clause tools - `pacgate_clause_risk`, `pacgate_clause_flags`, `pacgate_route_language` - which forward to the optional `laya` service. laya is a local System 1 decision engine: it answers typed questions about a single clause (how risky, which risk flags, which party it favors) in one forward pass and returns calibrated probabilities rather than generated prose. There is no API key and no cloud call, which is the reason it was chosen over a metered judgment API.

The server also exposes `pacgate_web_search` and `pacgate_web_scrape`, which call the Firecrawl API for current events, company news, and public pages the legal databases do not cover. They are inert until `FIRECRAWL_API_KEY` is set in `.env`; with no key they answer `available=false` and every other tool is unaffected. Results are graded `auxiliary_db` - a web page is never authority.

Firecrawl is deliberately NOT a fourth MCP server in `deer-flow-extensions-config.json`. deer-flow's `get_mcp_tools()` gathers one load task per configured server with `asyncio.gather` and no `return_exceptions`, then returns `[]` on any exception. One unreachable remote server would therefore remove pacgate, OpenViking and officecli from the agent at the same time, silently. Keeping the calls on pacgate-mcp - a service the agent needs anyway - confines a web failure to two tools.

It is off by default. The two checkpoints are about 1.45 GB on disk, so a machine that does not run it is not broken: the three tools return `available: false` with a reason, and every other tool is unaffected. From 0.2.13 the checkpoints ship inside the image, so enabling it is one command: `docker compose --profile laya up -d laya`. There is nothing to stage and nothing to download at run time - the weights are checksum-verified when the image is built, and the service runs with `HF_HUB_OFFLINE=1`. Budget 4.4 GiB of host RAM for it: measured idle with both checkpoints loaded, because each keeps its own fp32 copy. That is host memory, not VRAM, and on a 16 GiB machine with a large local model already resident it is worth checking free memory first.

Both checkpoints are required. laya routes by script - Latin text to `english`, Chinese to `multilingual` - and a missing checkpoint makes that language fail with HTTP 500. Measured on English clauses, the `english` checkpoint is also the more accurate of the two. 3. QM → OpenViking: the QM core has `OPENVIKING_URL=http://openviking:1933` and `OPENVIKING_API_KEY` (the root key). The `pacgate-qm` sandbox tool (`ov-remember` / `ov-search` / `ov-read`) calls OpenViking from inside the agent sandbox at `http://host.docker.internal:1933` with `OPENVIKING_ACCOUNT=pacgate-law` + `OPENVIKING_USER`. 4. QM → pacgate: the `pacgate-qm` sandbox tool logs into pacgate-api (`PACGATE_API_EMAIL` / `PACGATE_API_PASSWORD`) to discover workflows, bind a QM scope to a matter, read/write matter memory, and execute workflows.

Verify the full loop

```
# 1. OpenViking MCP responds (root key from .env)
$key = (Get-Content .env | Select-String '^OPENVIKING_ROOT_API_KEY=').Line.Split('=')[1]
$body = '{"jsonrpc":"2.0","id":1,"method":"tools/list"}'
curl.exe -s -X POST http://localhost:1933/mcp -H "X-API-Key: $key" -H "Content-Type: application/json" -H "A
# Expected: tools find/search/read/remember

# 2. QM core reaches openviking + pacgate-api + Ollama
docker exec qm-pacgate-core sh -c "curl -s -o /dev/null -w 'openviking:%{http_code}\n' http://openviking:193

# 3. QM sign-in works (dev/cookie mode)
$body = '{"user":"admin@pacgate-law.com"}'
curl.exe -s -X POST http://localhost:8182/signin -H "Content-Type: application/json" -d $body
# Expected: {"ok":true,"user":"admin@pacgate-law.com"}
```

QM ↔ bundle network join (idempotent)

QM core is joined to the bundle network so it can resolve `openviking` and `pacgate-api` by name. If a `qm up`/`qm down` cycle drops the join, re-apply it:

```
docker network connect pacgate-ai-bundle_default qm-pacgate-core
```

> Note: QM is intentionally not a Docker Compose service in the bundle. It is managed by the `@yc-software/qm` CLI (`qm up` / `qm down`) with its own lifecycle. The network join is the only coupling — keep it that way. Do not rewrite QM as compose services; that fights the qm CLI and would recreate the containers on every `qm up`.

Stage 6: Smoke test checklist (both machines)

Run this checklist on each AIPC independently.

Core stack

- [ ] `docker compose -f compose.prod.yaml ps` shows 5 services up (incl. openviking)
- [ ] `curl http://localhost:8089/version` returns the release version + revision JSON
- [ ] `curl http://localhost:8089/pacgate/health` returns `ok` (NOT `/health` at the root - see Stage 2)
- [ ] `curl http://localhost:1933/health` returns healthy JSON
- [ ] Postgres has the `pacgate-law` tenant
- [ ] Admin user can log in at `http://localhost:8089/api/auth/login`
- [ ] deer-flow returns a real research response at `http://localhost:8089/research/`

qm collaboration

- [ ] `npm exec qm -- status` shows qm running
- [ ] `http://localhost:8182` loads the qm web UI
- [ ] Admin can sign in
- [ ] qm can list Pacgate workflow categories
- [ ] qm can execute one Pacgate workflow through the bridge

Ollama

- [ ] `ollama list` shows the required models

- [ ] `ollama ps` shows a GPU, not `100% CPU` - run `scripts/test-ollama-gpu-residency.ps1`, it must exit 0
- [ ] deer-flow can call Ollama for inference
- [ ] qm can call Ollama for inference

> The CPU-only trap - read this before you conclude a machine is slow.
>
> Ollama enumerates compute devices once, at server start, and it unpacks the ROCm/Vulkan backend into `lib\ollama\<vendor>_*` only after that. A server started before the unpack completes decides "CPU only" and keeps that decision for its entire lifetime - with a perfectly healthy install, a supported GPU, and no error anywhere.
>
> Measured 2026-11-04 on the RX 7600 XT box:
>
> That machine ran every local model on the CPU for 3.5 hours. One "Reply with exactly: OK" took 377 s; the same request after restarting Ollama took 7.4 s at `100% GPU`. A chat turn looked like a hang while `smoke-full-stack` reported every lane green.
>
> First action if local inference feels slow or the agent appears to stall:
>
> ```
> powershell ollama ps # PROCESSOR must say GPU, not 100% CPU
> .\scripts\test-ollama-gpu-residency.ps1
> ```
>
> If either says CPU, restart the Ollama app (not the model) and re-check. Do not reinstall ROCm, do not replace the model, and do not touch the stack - the install is almost certainly fine.
>
> This is also why the residency gate is part of `run-all-checks.ps1`: a fresh AIPC can lose this same race on its very first start.

Data

- [ ] `./data/tenants/` directory exists and is writable
- [ ] `./openviking/` directory exists and persists across restarts
- [ ] Document upload works through the API
- [ ] Matter memory persists after a deer-flow research run
- [ ] Cross-session recall: a fact stored via OpenViking `remember` is recalled via `search` in a later session

Managing the stack after deployment

Start and stop

```
# Start core stack
docker compose -f compose.prod.yaml up -d

# Stop core stack
docker compose -f compose.prod.yaml down

# Start qm
cd C:\pacgate-ai-pr\deploy\client-bundle\qm-pacgate
docker compose -f compose.qm.yaml up -d

# Stop qm (NO -v: .env and the volumes stay)
docker compose -f compose.qm.yaml down
```

> `qm up` / `qm down` cannot run on Windows (the CLI's `which()` shells `/bin/sh` → ENOENT); compose is the only working qm lifecycle on the AIPC and it carries the patches that keep model routing and the sandbox lane durable.

Update to a new version

```
cd C:\pacgate-ai-pr\deploy\client-bundle
.\install.ps1 -Update
```

`-Update` now does all of the following, so a separate `git pull` is no longer needed (it was the most easily forgotten step, and forgetting it silently ran new images against old config):

Step	What it does
1	Refreshes the repo working tree (fast-forward only)
2	Re-renders the MCP config from the template, backing up any change
3	Pulls new GHCR images
4	Restarts the stack, and reloads nginx

It refuses, rather than guessing, when the machine has local work:

- uncommitted changes in the repo -> the repo update is skipped, and the changed files are listed. Nothing is stashed, reset, or discarded.
- local commits the remote does not have -> it will not merge or rebase. The repo stays as-is and the rest of the update continues.

Use `-SkipRepoPull` if you deliberately want an image-only update:

```
.\install.ps1 -Update -SkipRepoPull
```

qm is reported, not auto-updated

If qm is running on the machine, `-Update` also checks whether its sandbox image still matches its source:

```
[OK] qm sandbox matches its source (38e062ec7c5c...)
```

or

```
[WARN] qm sandbox source has CHANGED since the image was pinned.
       qm is running OLD skills and tools. Rebuild + repin:
         cd deploy\qm-pacgate
         npm exec qm -- sandbox build   # then repin the printed digest
         pwsh -File ..\..\scripts\qm-sandbox-fingerprint.ps1 -Write
```

This matters because qm's agent executes inside a sandbox image pinned by digest in `deploy/qm-pacgate/qm.config.jsonc`. Digest pinning is correct - the isolation boundary should be immutable - but it means a repo update can change `deploy/qm-pacgate/sandbox/` while the image stays exactly as it was. The agent then keeps running the old skills and tools, with no error anywhere.

`-Update` reports this rather than rebuilding on its own, because the rebuild needs Node 24 + npm + buildx and the digest must be repinned afterwards - a config change we should not make unattended on a client machine. A wrong automatic rebuild is a worse failure than a visible warning.

Data is preserved across an update: - `./data/tenants/` (volume mount) - matters, documents, memory - Postgres data (named volume) - metadata database

Is this machine current? Ask it.

```
curl.exe -s http://localhost:8089/version
```

```
{"version":"<release>","revision":"<git sha>"}
```

This reports the version compiled into the running pacgate-api binary, and the commit it was built from. It deliberately does not echo the compose pin or the image tag: those record what was deployed, and the failure worth catching is exactly

the case where the deployed artifact and the running process disagree.

Before this existed there was no way to tell a current machine from one behind -both looked identical from the outside, and the only check was to SSH in and read the compose file, hoping the containers matched it.

After a security fix: check for documents sanitized by an old build

Some fixes cannot repair existing data by shipping an image, and the sanitizer change in 0.1.31 is one of them. Builds up to and including 0.1.30 stamped a chunk `sanitized` while leaving the raw text in the row, and the retrieval gate then served that text to chat. Upgrading the image does not rewrite those rows.

So on any machine that ever ran 0.1.28, 0.1.29 or 0.1.30, run the audit after updating:

```
pwsh -File C:\pacgate-ai-pr\scripts\audit-sanitized-chunk-egress.ps1
```

It is read-only and touches nothing. A clean machine prints:

```
== RESULT: no raw Tier-1 match found in a sanitized chunk ==
```

Any other result names the affected documents:

```
== RESULT: 106 raw match(es) inside SANITIZED chunks - LEAK PRESENT ==
affected documents (up to 25):
  matter=56723800-...  doc=820f4904-...  name=leak
```

For each affected document, re-run its sanitize job. The job now replaces the stale chunk text with the redacted rendering, which repairs the row and keeps retrieval coverage. There is a script that does this for every affected document in one pass:

```
# See what would be repaired first - read-only, changes nothing.
pwsh -File C:\pacgate-ai-pr\scripts\remediate-sanitized-chunk-egress.ps1 -WhatIf

# Repair them.
pwsh -File C:\pacgate-ai-pr\scripts\remediate-sanitized-chunk-egress.ps1
```

It logs in as `PACGATE_API_EMAIL` using `PACGATE_API_PASSWORD` from `deploy\client-bundle\.env`, so it needs real credentials - it exits 2 and says so rather than half-working if the password is still the placeholder.

The manual equivalent, if you prefer one document at a time:

```
$login = Invoke-RestMethod -Uri "http://localhost:8089/pacgate/api/auth/login" `
  -Method Post -ContentType 'application/json' `
  -Body '{"email":"admin@pacgate-law.com","password":"<password>"}'
$hdr = @{ Authorization = "Bearer $($login.token)" }
Invoke-RestMethod -Uri "http://localhost:8089/pacgate/api/documents/<doc-id>/sanitize" `
  -Method Post -Headers $hdr -ContentType 'application/json' -Body '{"data_level":"T3"}'
```

Then re-run the audit and confirm the document no longer appears. Do not skip this step: the repair reports what it did, the audit measures what is true, and on the first machine those two disagreed - twice - which is how two defects in the audit itself were found.

Order matters: update to 0.1.31 FIRST. On 0.1.30 and earlier the sanitize job has no rewrite path at all, so re-running it changes nothing and the audit still reports every affected document.

Two things worth knowing before starting:

- The repair re-embeds the document. It rewrites every chunk, so a large document makes one embedding call per chunk. Correct, but proportional to document size rather than to how much actually leaked.
- Deleting the rows instead is also safe, and faster, but loses the document's RAG coverage until it is re-sanitized. Prefer the re-run.

Neither route works on 0.1.30 or earlier: that code has no rewrite path at all, so re-running the job there changes nothing. Update first.

Switch models

deer-flow (research workspace): 1. Edit `deer-flow-config.yaml` - reorder the `models` list (first entry = default) 2. Restart: `docker compose -f compose.prod.yaml restart deer-flow`

qm (co-working workspace): 1. Edit `deploy/client-bundle/qm-pacgate/qm.config.jsonc` if the model set changes 2. Recreate core (compose keeps the patch mounts, so the routing stays durable): `powershell cd C:\pacgate-ai-pr\deploy\client-bundle\qm-pacgate docker compose -f compose.qm.yaml up -d --force-recreate core`

Register new users

`POST /api/auth/register` is first-user-only and returns 403 after the bootstrap account (see Stage 3). Create every later account through the admin route:

```
# /pacgate prefix required - see the note in Stage 3. Bearer = admin's token
# from POST /pacgate/api/auth/login (see Stage 3 for the login step).
$body = @{email="<user>@pacgate-law.com"; password="<password>"} | ConvertTo-Json
Invoke-RestMethod -Uri "http://localhost:8089/pacgate/api/auth/users" -Method POST `
  -Headers @{Authorization="Bearer <admin-token>"} -Body $body -ContentType "application/json"
```

Backup the database

```
docker exec pacgate-db pg_dump -U pacgate pacgate > backup.sql
```

Check logs

```
docker compose -f compose.prod.yaml logs -f pacgate-api
docker compose -f compose.prod.yaml logs -f deer-flow
```

Known limitations

- `qm up` / `qm down` do not work on Windows. The qm CLI's docker lifecycle is POSIX-only: its `which()` runs `execFileSync("/bin/sh", …)`, which is `ENOENT` on every Windows AIPC (measured 2026-10-06). Use the compose path above (`docker compose -f compose.qm.yaml up -d`) - same topology, same container names, and it carries the Pacgate patches, so nothing is lost across recreates.
- Each machine has its own independent Postgres and `./data/tenants/` directory. Matter data is not shared between machines unless you later add a private mesh and a sync or single-authority model.
- The PkuLaw connector token is expired. Regenerate it at `https://mcp.pkulaw.com` and set `PKULAW_API_KEY` in `.env` if China-law search is needed during the pilot.
- The Firecrawl web lane is off until a key is pasted. Set `FIRECRAWL_API_KEY` in `deploy/client-bundle/.env` and restart; one value arms the `/api/search` connector, the agent's `pacgate_web_search` / `pacgate_web_scrape` tools, and (via `setup-qm.ps1`) the QM sandbox's `firecrawl-qm` CLI. Verify with `curl -s http://localhost:8089/pacgate/api/search/health` - the `firecrawl` row flips to `available=true`. With no key, all three surfaces report "not configured" and nothing else changes.
- Four WASM crates (citation-check, clause-parser, doc-validator, rule-engine) remain stubs. These are future-blueprint work and do not affect Phase 1 pilot functionality.
- Model selection: the API defaults to models that may not exist on the target machine. After Stage 3, apply per-tenant model overrides so the LLM tiers point at models actually present in `ollama list` on that machine. Recommended pilot set (benchmarked 2026-08-28): `gemma4:12b-it-qat` (Main — 13s/tool-round, schema-valid tool calls, verified end-to-end), `qwen3.8:27b-mtp-q4_K_M` (Mid — 73s/tool-round, stronger quality for batch tabular review), `nomic-embed-text:latest` (embeddings). Avoid reasoning-mode models (e.g. nemotron) for interactive tiers — they can hang long docx generations.
- Large PDFs need the outline lane to work. Uploaded PDFs are converted to Markdown so the agent can read them. The image prefers `pymupdf4llm` and falls back to MarkItDown; only `pymupdf4llm` reproduces a PDF's

heading hierarchy. If the fallback runs, the agent is told "No structural headings detected", gets no section map, and a long document becomes unnavigable — it then re-reads the same file turn after turn until the run hits its step ceiling. This is a real observed failure mode (2026-11-06), not a cosmetic issue. Verify after any image rebuild with:

```
powershell pwsh -File scripts/test-pdf-outline-extraction.ps1
```

It exits 1 if the converter is missing, if the deployed config routes around it, or if a structured PDF yields no outline.

Files referenced

File	Purpose
deploy/client-bundle/compose.prod.yaml	Docker Compose for pacgate-api + deer-flow + Postgres + nginx
deploy/client-bundle/install.ps1	One-click Windows installer for the core stack
deploy/client-bundle/setup-qm.ps1	qm bootstrap script (secrets, config, sandbox build)
deploy/client-bundle/.env.example	Template for client secrets
deploy/client-bundle/ollama-models.txt	Models to pre-pull
deploy/client-bundle/deer-flow-config.yaml	Multi-model deer-flow config (5 models, switchable)
deploy/pacgate-pins/pymupdf4llm.constraints.txt	Pins the PDF→Markdown converter pair the deer-flow image installs
scripts/test-pdf-outline-extraction.ps1	Gate: a structured upload must yield a document outline the agent can navigate
deploy/qm-pacgate/qm.config.jsonc	qm local deployment config
deploy/SETUP-AND-OPERATIONS.md	Full 3-day on-site install guide (reference)
deploy/DEPLOYMENT-GUIDE.md	Engineer-level deployment details (reference)